PyCIL: A Python Toolbox for Class-Incremental Learning

Da-Wei Zhou, Fu-Yun Wang, Han-Jia Ye and De-Chuan Zhan

Nanjing University

zhoudw@lamda.nju.edu.cn wangfuyun@smail.nju.edu.cn

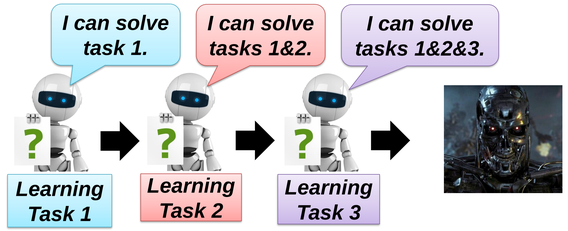
arXiv.org Papers With Code

arXiv:2112.12533 2021.

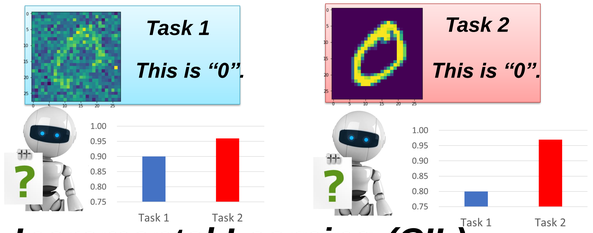
Highlights

- *We propose a Python toolbox that implements several key algorithms for class-incremental learning.*
- *It contains a number of founding works but also provides current state-of-the-art algorithms.*
- *We made a detailed summary and comparison.*

What people think about AI:

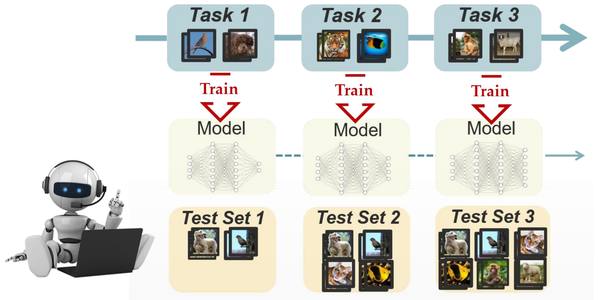


The real situation is: Catastrophic Forgetting!

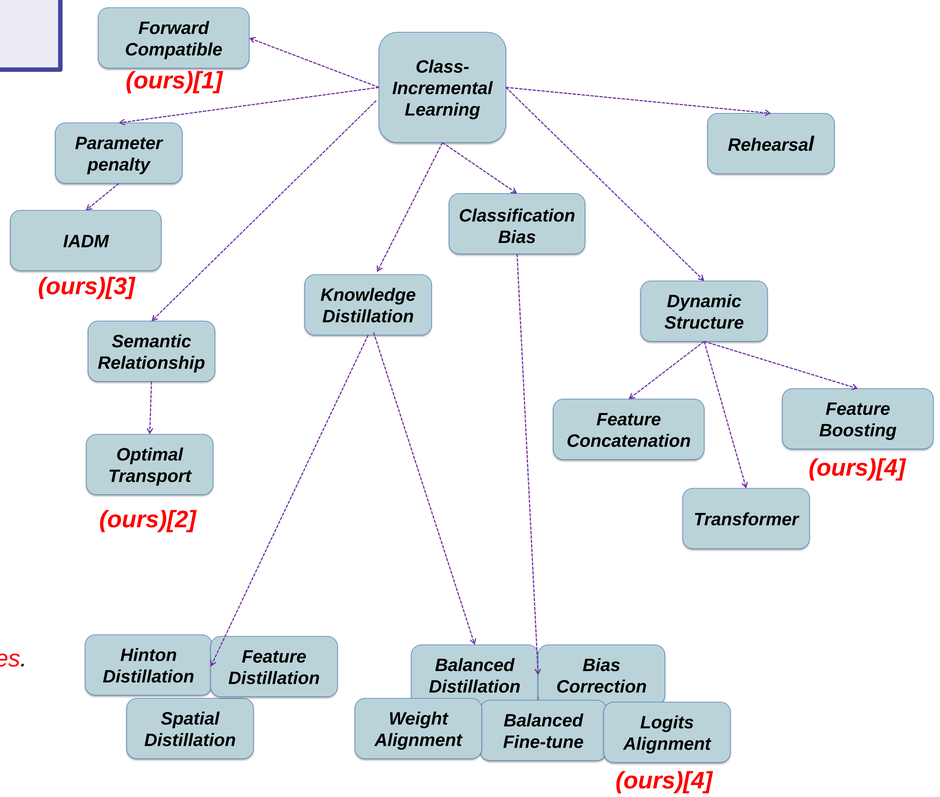


Class-Incremental Learning (CIL)

learning novel categories while maintaining performance on old ones.



Methods Tree



[1] FACT: Forward Compatible Few-Shot Class-Incremental Learning. ***CVPR2022.***

[2] Coil: Co-Transport for Class-Incremental Learning. ***ACM MM'21.***

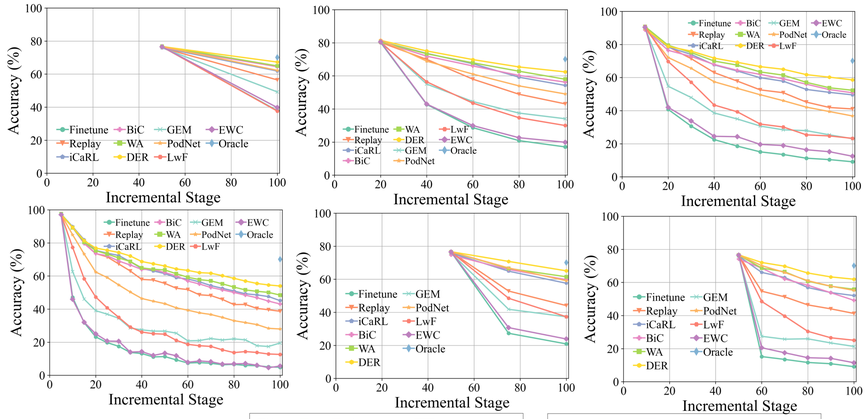
[3] IADM: Adaptive Deep Models for Incremental Learning: Considering Capacity Scalability and Sustainability. ***KDD19.***

[4] FOSTER: Feature Boosting and Compression for Class-Incremental Learning. ***arXiv2022.***

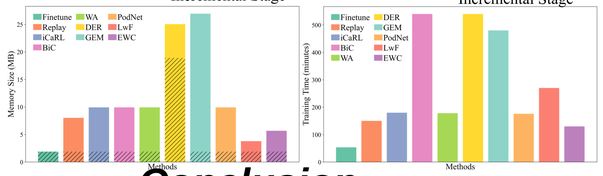
Methods Comparison

Benchmark results

- ***CIFAR100****: B0 with 5, 10, 20 steps; B50 with 2, 5, 10, 25 steps.*
- ***ImageNet100****: B0 with 5, 10, 20 steps; B50 with 2, 5, 10, 25 steps.*
- ***ImageNet1000****: B0 with 10 steps.*



Storage and computation overhead



Conclusion

We propose a python toolbox PyCIL for CIL to ease the burden of CIL researchers. The toolbox provides implementation of both founding and state-of-the-art works.

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